Project Status Update: Minimum-Energy Trajectory Optimization for Lunar Rover Navigation

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1. Dynamic Model

The chosen energy-aware nonlinear dynamic rover model is based on the formulation in Hu et al. (2025). The model includes the rover kinematics, translational power, rotational power, resistive power, baseline subsystem power, total consumed power, available power, instantaneous power feasibility evaluation, and cumulative energy computation. Resistive forces of the dynamic model depend on empirical coefficients dependent on the terrain, not described in Hu et al. (2025). We propose to estimate the missing parameters following terramechanics modeling Li and Bingham (2022). The model has already been implemented in code, assuming no resistance coefficients for now. Fig. 1 shows a simulation of rover dynamics going around a circle until battery drains.

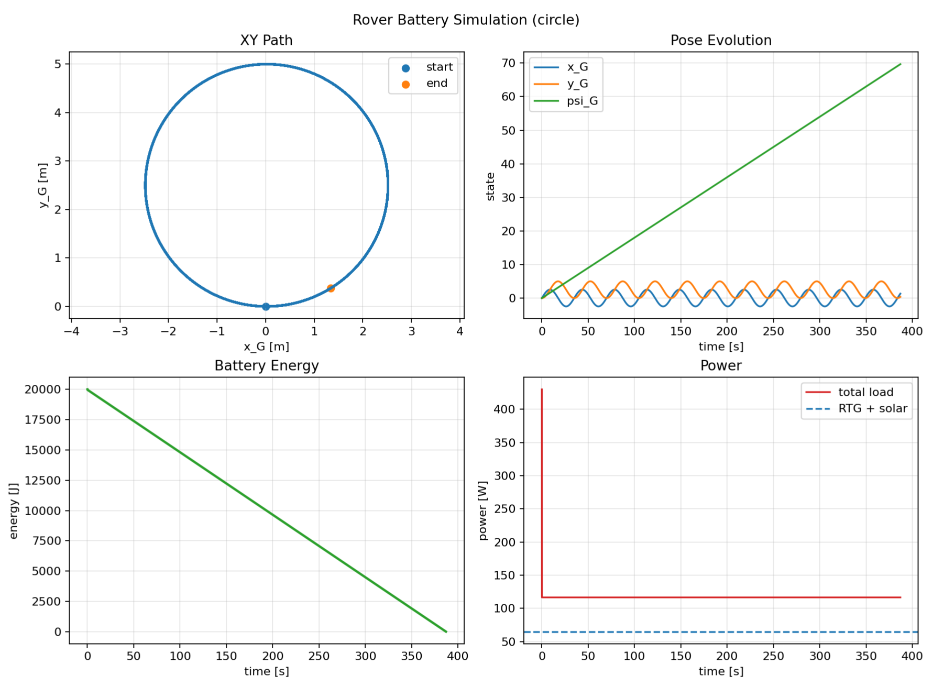

Figure 1: Rover Dynamics Simulation

2. Trajectory Optimization

The trajectory optimization objective is similar to Hu et al. (2025).

$$\begin{aligned}
\min_{c,T_i} \quad & \int_0^{T_f} c_T \sum_i T_i + c_r \mathcal{R}(L_G) + J_P \\
\text{s.t.} \quad & p(0) = p_0, \\
& p(T_f) = p_g, \\
& \dot{x}_B(t) \leq \dot{x}_{B,\max}, \\
& |\ddot{x}_B(t)| \leq \ddot{x}_{B,\max}, \\
& |\ddot{y}_B(t)| \leq \ddot{y}_{B,\max}, \\
& P_{\text{cons}}(t) \leq P_{\text{avail}}(t), \qquad \forall t \in [0, T_f].
\end{aligned}$$

Here, c_T is the weight on traversal time, and J_P is a soft penalty for power constraint violations. The power margin is defined as

$$\tau(t) = P_{\text{avail}}(t) - P_{\text{cons}}(t) \tag{1}$$

And the soft power penalty is

$$J_P = c_P \int_0^{T_f} \text{softplus}\left(-\tau(t)\right)\, dt \tag{2}$$

where c_P is the power penalty weight, which penalizes states where consumed power exceeds available power.

2.1 Optimization Method

Sequential Convex Programming will be used for trajectory generation, a well known optimization method with deep roots in aerospace applications Malyuta et al. (2022).

3. Terrain Risk Field and Traversability

The objective in the previous section depends on the continuous terrain-risk term $c_r \mathcal{R}(L_G)$, and rover stability imposes a hard slope limit of 20°. We implement Stages 1–2 of the proposal pipeline as a region-agnostic terrain module.

Terrain source. We generate a physically-motivated synthetic complex-crater DEM — a flat floor, cosine-profile wall, Gaussian rim, exponential ejecta blanket, and spatially correlated micro-roughness — at 118 m/px on a 30 km cutout. The IAU gazetteer is used to identify Artemis-relevant south-pole craters (Shackleton, de Gerlache, Faustini, Cabeus, Nobile, and others) for later use with real LOLA data.

Derived layers. Slope is computed from the elevation gradient via central differences with metric spacing, and roughness as the local elevation standard deviation over a small window. The traversability mask marks cells below the slope and roughness limits as drivable, forming the obstacle set for the Stage 3 global planner. The continuous risk field combines a quadratic slope penalty that activates beyond a soft 12° threshold, a quadratic roughness penalty, and a large constant penalty on non-traversable cells, with a final Gaussian smoothing so the risk gradient is well-behaved inside SCP.

Optimizer interface. The module exposes bilinear risk sampling in continuous metric coordinates, a central-difference risk gradient, and a trapezoid-rule path integral matching the objective term above, along with the boolean obstacle set.

Validation. On a 30 km cutout containing a 22 km crater, elevation spans −702 m to +291 m, peak slope reaches 25.6°, and 73.3% of the area is traversable (plains and floor drivable, only the wall/rim annulus excluded). Risk separates by roughly four orders of magnitude between the crater floor and the wall. As an end-to-end check, a 26 km straight-through traversal scores about 640× higher than a longer semicircular detour around the rim, confirming the risk field correctly prefers the safe route over the short but hazardous one (Fig. 2).

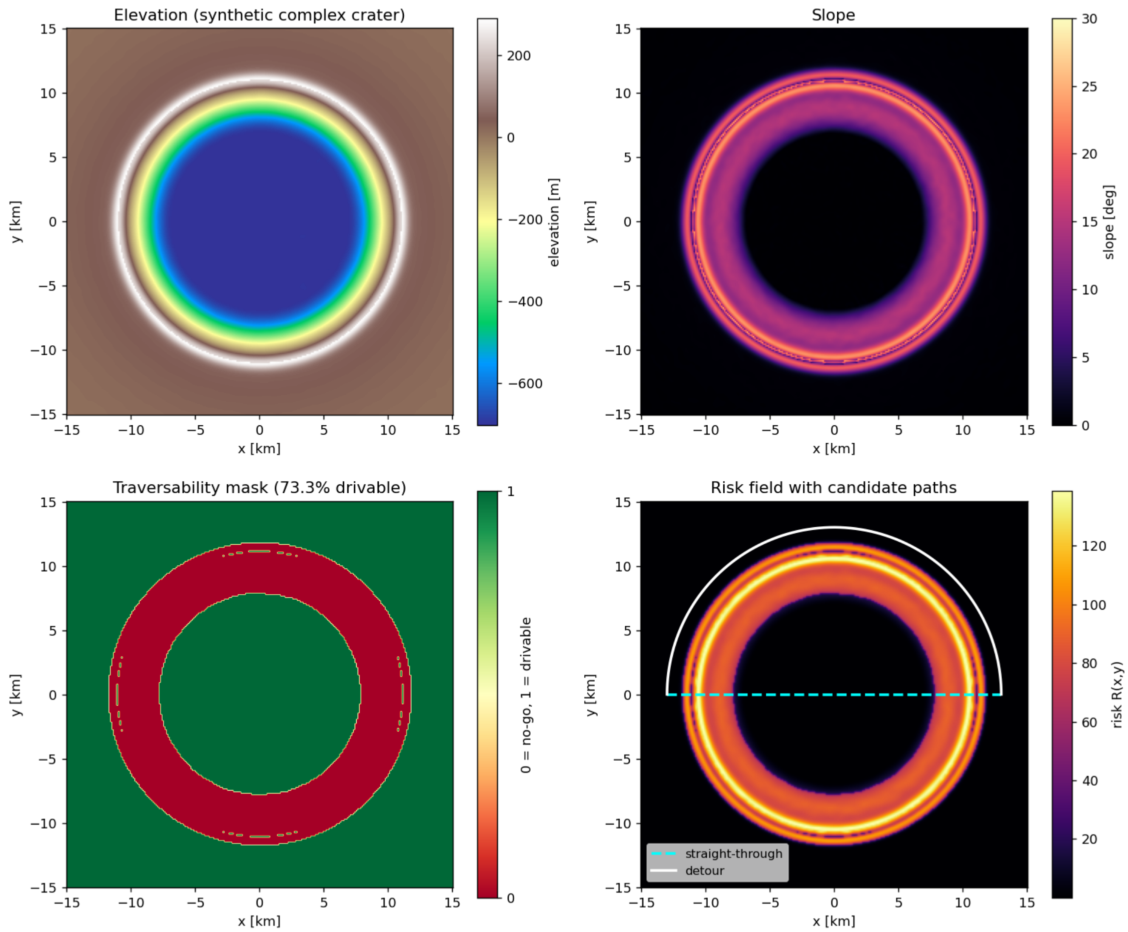

Figure 2: Terrain pipeline validation. Top-left: synthetic complex-crater elevation (flat floor, cosine wall, Gaussian rim, exponential ejecta). Top-right: slope. Bottom-left: boolean traversability mask (73.3% drivable). Bottom-right: smooth risk field $\mathcal{R}(x, y)$ with the straight-through (cyan, dashed) and detour (white) candidate paths overlaid — the detour stays in the dark plain.

4. Next Steps and Revised Timeline

Next steps include finishing implementing the terramechanics model for resistive forces, and implementation of the SCP-based trajectory optimizer. The latter includes implementing the convex subproblem, adding trust-region or regularization terms, enforcing the linearized dynamics and constraints, and iterating until the trajectory converges. With the terrain pipeline now producing risk samples, risk gradients, a path-integral risk, and the boolean obstacle set, the SCP subproblem can be instantiated and warm-started from an A*/Hybrid A* solution on the traversability mask. Revised timeline with next steps can be found in Table 4

Step	Week
Revised Dynamic Model	8
Path Planner	8
SCP	8
Report	9

References

Tianxin Hu, Weixiang Guo, Ruimeng Liu, Xinhang Xu, Rui Qian, Jinyu Chen, Shenghai Yuan, and Lihua Xie. Energy-constrained navigation for planetary rovers under hybrid rtg-solar power, 2025. URL `https://arxiv.org/abs/2509.15062`.

Zu Qun Li and Lee K. Bingham. NASA White Paper: Terramechanics for LTV Modeling and Simulation. White Paper 20220010732, NASA Johnson Space Center, July 2022. URL `https://ntrs.nasa.gov/citations/20220010732`. NASA Technical Reports Server (NTRS).

Danylo Malyuta, Taylor P. Reynolds, Michael Szmuk, Thomas Lew, Riccardo Bonalli, Marco Pavone, and Behçet Açıkmeşe. Convex optimization for trajectory generation: A tutorial on generating dynamically feasible trajectories reliably and efficiently. *IEEE Control Systems Magazine*, 42(5):40–113, 2022. doi: 10.1109/MCS.2022.3187542.